DRY PIPE VALVE ACCELERATOR

MODEL F311 POSI-QUIK®

GENERAL DESCRIPTION

The Model F311 Accelerator (Ref. Figure A) is a quick opening device, which can be attached to 4 and 6 inch Model F302 and F3021 Dry Pipe Valves, for the purpose of reducing the time to valve operation following the operation of one or more automatic sprinklers.

The Accelerator automatically adjusts to both small and slow changes in system pressure but trips when there is a rapid and steady drop in pressure (such as will occur when a sprinkler operates). Upon tripping, the Accelerator transmits system air pressure to the intermediate chamber of the dry pipe valve. This neutralizes the differential pressure holding the valve closed and permits its waterway clapper to open due to the force exerted by the supply pressure.

The F311 has a unique, positive action, internal anti-flood device and a ball float which combine to prevent water and water borne debris from entering the more sensitive operating areas of the accelerator.

The anti-flood device seals and latches immediately upon accelerator trip, due to a sprinkler operation, without having to wait for a pressure build-up in the intermediate chamber of the dry pipe valve. The latching feature keeps the anti-flood device sealed, even while the system is being drained.

The ball float seals the pilot chamber inlet port if there is an accidental trip of the dry pipe valve due, for example, to an air compressor failure combined with a slow loss in system air pressure due to a leak.

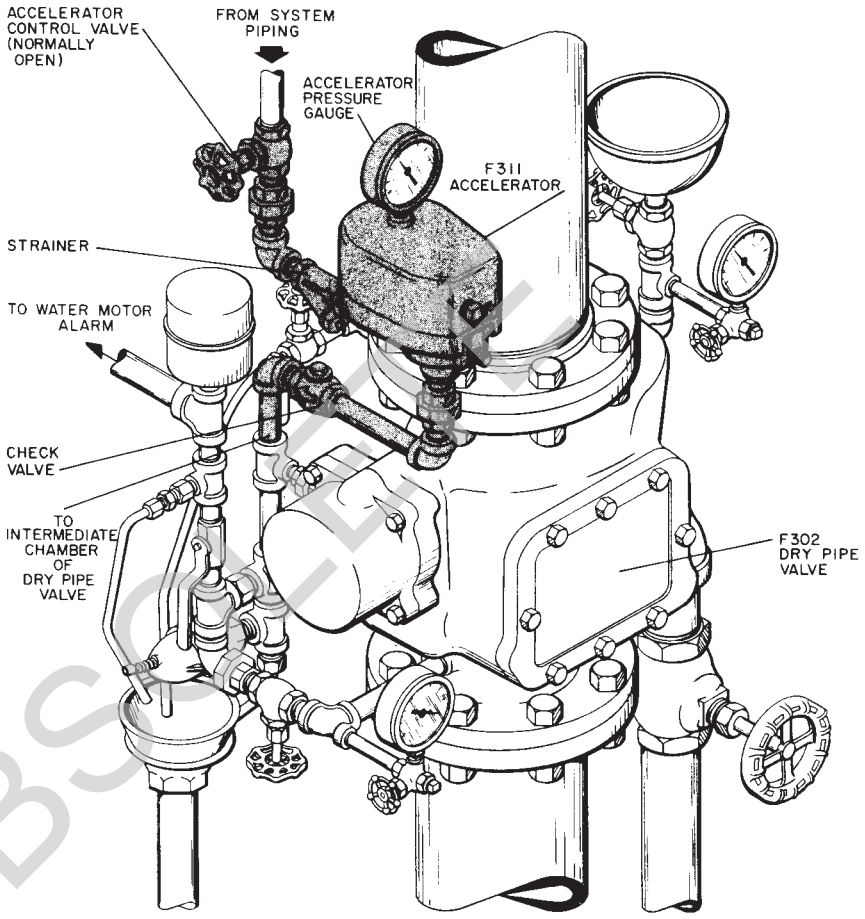


FIGURE A
MODEL F311 ACCELERATOR ATTACHED TO A MODEL F302 DRY PIPE VALVE

APPROVALS AND STANDARDS

The Model F311 Accelerator is listed by Underwriters Laboratories Inc. and by Underwriters' Laboratories of Canada. It is approved by Factory Mutual Research Corporation and the Loss Prevention Council.

The Model F311 Accelerator is approved by the New York City Board of Standards and Appeals under Calendar Number 236-79-SA.

The above listings and approvals of the F311 only apply to its use with the 4 and 6 inch Model F302 and Model F3021 Dry Pipe Valves.

The Model F311 Accelerator is also approved by the Verband der Schadenversicherer for use with the DN100 and DN150 Model F302DIN Dry Pipe Valves.

WARNING

The Model F311 Accelerator described herein must be installed and maintained in compliance with this document, as well as with the applicable standards of the National Fire Protection Association, in addition to the standards of any other authorities having jurisdiction. Failure to do so may impair the integrity of this device.

The owner is responsible for maintaining their fire protection system and devices in proper operating condition. The installing contractor or manufacturer should be contacted relative to any questions.

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*Assume quick opening of a 1/2" Inspector"s Test Connection.

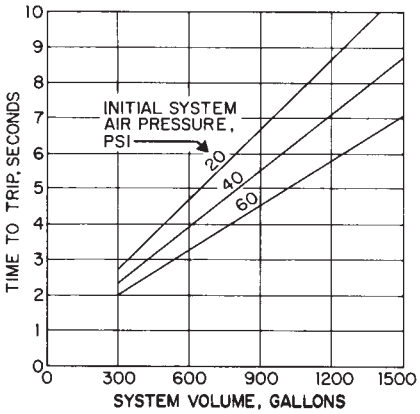


FIGURE B
TYPICAL OPERATING TIMES FOR F311 ACCELERATOR*

TECHNICAL DATA

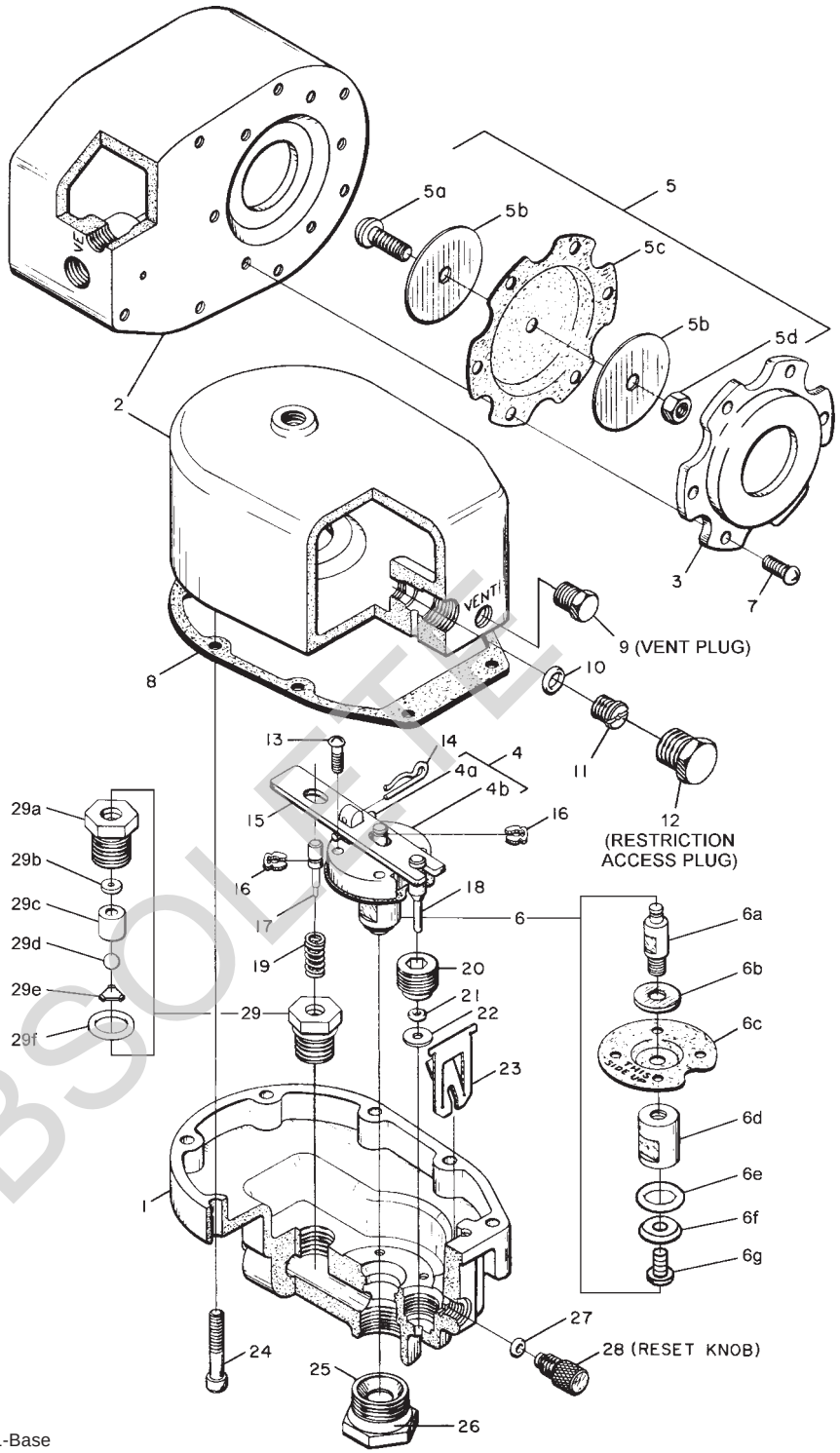
The Model F311 Accelerator is rated for use at a maximum water supply pressure of 175 psi and a maximum system air (or nitrogen) pressure of 60 psi. It is a light weight unit which features a small volume Differential Chamber for quick filling, a sintered metal Restriction that provides dependable high-sensitivity, an internal anti-flood device that seals immediately upon tripping, a ball float, and self-draining of the connection to the sprinkler system piping.

The F311 is designed to trip when the system air pressure drops at a rate exceeding about 1 psi/minute. A graph of the typical times to Accelerator trip versus system volume is given in Figure B, for initial system air pressures of 20, 40 and 60 psi. The time to trip may be slightly longer than that indicated in Figure B for particularly complex or small diameter piping systems. Tripping of the dry pipe valve takes place immediately after the Accelerator trip.

NOTE

Quick operation of the Accelerator does not ensure that the fire protection system will meet the water delivery time requirement of the authority having jurisdiction (following opening of the Inspector's Test Connection). The sprinkler system designer needs to be aware of the fact that the water delivery time is primarily determined by the configuration and volume of the piping network, system air pressure at time of Accelerator trip, and water supply characteristics.

The Accelerator components are illustrated in Figure C. The Base, Cover, Upper Diaphragm Plate, and Pivot Plate are cast from aluminum alloy

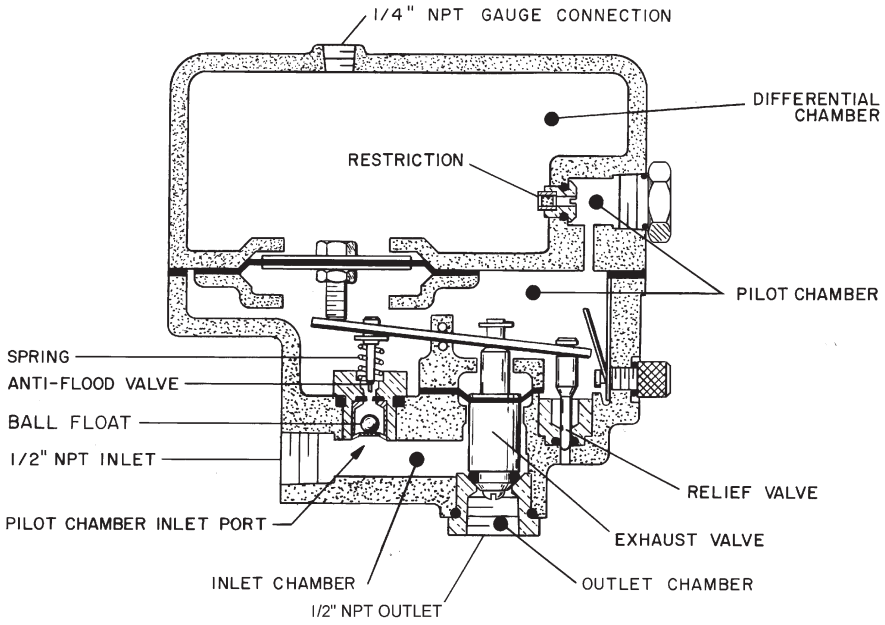


1-Base
2-Cover
3-Upper Diaphragm Plate
4-Pivot Plate Assembly
a-Spirol Pin
b-Pivot Plate
5-Plunger
a-Pan Hd. Machine Screw
b-Upper Diaphragm Retaining Ring
c-Upper Diaphragm
d-Jam Nut
6-Exhaust Valve
a-Upper Plug
b-Washer
c-Lower Diaphragm
d-Lower Plug
e-O-Ring*
7-Rd. Hd. Machine Screw, 1/4"-20 UNC x 5/8" (6 req'd)
8-Cover Gasket
9-Vent Plug
10-O-Ring*
11-Restriction
12-Restriction Access Plug
13-Pan Hd. Machine Screw No. 10-32 UNF x 5/8" (4 req'd)
14-Cotter Pin
15-Lever
16-Retaining Ring
17-Anti-Flood Valve
18-Relief Valve
19-Spring
20-Relief Valve Seat
21-O-Ring*
22-Seal Washer
23-Latch
24-Fillister Hd. Machine Screw 1/4"-20 UNC x 1-1/2" (8 req'd)
25-Plug Seat
26-O-Ring*
27-O-Ring*
28-Reset Knob
29-Anti-Flood Seat Assembly w/Ball Float
a-Insert
b-Seal
c-Guide
d-Ball
e-Clip
f-O-Ring*

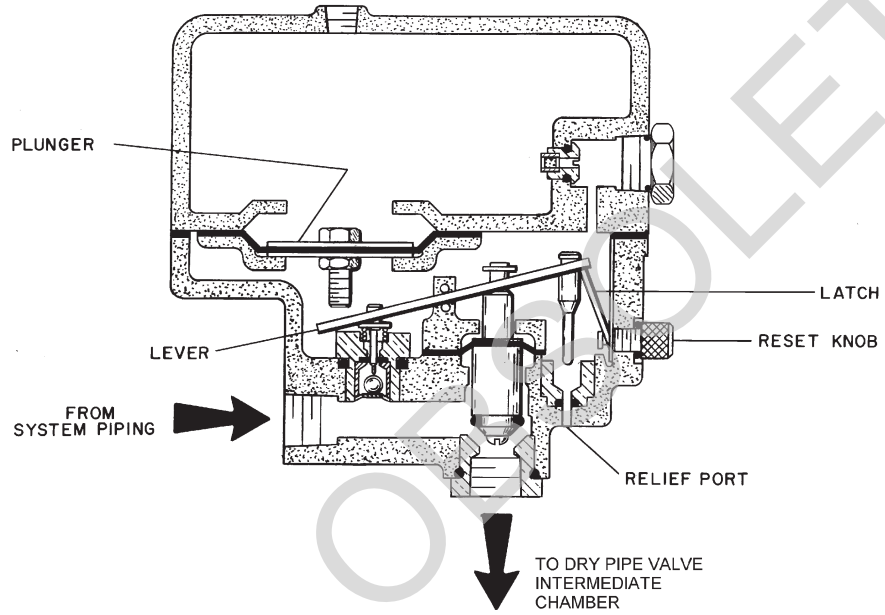
* Requires thin film of fluorosilicone grease

FIGURE C
MODEL F311 ACCELERATOR ASSEMBLY

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Accelerator In Set Position



Accelerator In Trip Position

FIGURE D
MODEL F311 ACCELERATOR IN SET AND TRIPPED POSITIONS

A356.0-T6 (per ASTM B26 or B108) and they are Alodine coated after machining per MIL-C-5541C. The exterior surfaces of the Base and Cover are also painted red. All other metallic components are fabricated from an austenitic series stainless steel; and the Restriction is a sintered stainless steel.

The Upper Diaphragm, Lower Diaphragm, and Pilot Chamber Inlet Seal are molded from an EPDM grade of rubber. The Cover Gasket is Neoprene, and the Ball Float is Polypropylene. The O-rings are molded from EPDM rubber except in the case of the Relief Port O-ring which is silicone.

The serial number and last two digits of the year of manufacture of the Accelerator are metal stamped on the right side of the Base.

OPERATION

The Inlet Chamber of the Accelerator (Ref. Figure D-Set Position) is pressurized via its connection to the system (at a point above the maximum expected level of priming plus drain back water). The Pilot Chamber is, in turn, pressurized through its inlet port which is formed by the annular opening around the lower tip of the Anti-Flood Valve. As the Pilot Chamber increases in pressure, the Differential Chamber is pressurized through the Restriction.

The Accelerator is in its Set position while it is being pressurized as well as after the Inlet, Pilot Chamber and Differential Chamber pressures have equalized. When in the Set position, the Outlet Chamber is sealed off by the Exhaust Valve which is held against its seat by a combination of the Spring pushing up against the Lever and the net downward force exerted by the pressure in the Pilot Chamber.

Both small and slow changes in system pressure are accommodated by flow through the Restriction. When, however, there is a rapid and steady drop in system (i.e., Inlet and Pilot Chamber) pressure, the pressure in the Differential Chamber reduces at a substantially lower rate. This condition creates a net downward force on the Plunger which rotates the Lever. As the Lever is rotated, the Relief Valve is raised out of the Relief Port and the Anti-Flood Valve is depressed downward into the Pilot Chamber Inlet Port, venting the Pilot Chamber.

The system pressure in the Inlet Chamber then forces (raises) the Exhaust Valve off its seat. This continues the rotation of the Lever into the Tripped (latched) position as shown in Figure D. As the Exhaust Valve is raised off its seat, system pressure is transmitted to the intermediate chamber of the dry pipe valve which neutralizes the differential pressure holding the valve closed.

Following the dry pipe valve trip, major water borne debris is prevented from entering the Accelerator (via the connection to the system piping) by the Strainer located at its Inlet. Water and any fine water borne debris such as silt is prevented from entering the Pilot Chamber by virtue of the Anti-Flood Valve having sealed off its inlet port. The Check Valve located downstream of the Accelerator Outlet prevents any water borne debris from entering the Accelerator via the connection to the intermediate chamber of the dry pipe valve.

After the Accelerator/dry pipe valve trip and the sprinkler system has been

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drained, the piping from the system to the Accelerator must also be drained and the Accelerator reset/inspected according to the instructions given in the Setting Procedure Section. The Accelerator is reset simply by unscrewing the knurled Reset Knob until it resists further turning and then screwing the Reset Knob back in until it is finger tight.

The rate-of-flow through the Restriction has been set such that the F311 Accelerator provides the maximum practical sensitivity to a loss in system pressure due to a sprinkler operation while still being capable of automatically compensating for normal variations in system pressure such as are caused by environmental temperature changes. A test for verifying that the rate-of-flow through the Restriction is within the range for optimum Accelerator performance is given in the Setting Procedure Section.

INSTALLATION

NOTES

Proper operation of the Model F311 Accelerator depends upon it being installed in accordance with the instructions given below. Failure to follow these instructions may prevent the device from functioning properly as well as void listings, approvals and the manufacturer's warranties.

The F311 Accelerator can be used with differential type dry pipe valves other than the F302 and F3021, provided that the Accelerator is installed and trimmed in accordance with the instructions given below.

The following are the installation instructions for the Model F311 Accelerator.

1. The Accelerator must be positioned vertically and trimmed per the arrangement shown in Figure E. Apply pipe thread sealant to male threads only.

2. The Strainer located at the Accelerator Inlet must be PSN 52-353-1-005 and it must be installed with its arrow pointed towards the Accelerator.

3. The swing type Check Valve located in the line between the Accelerator Outlet and the intermediate chamber of the dry pipe valve must be PSN 46-049-1-004 and it must be installed horizontally with its arrow pointed in the direction of flow to the intermediate chamber.

4. The connection to the system piping must be located so that drain back water will not flow into the Accelerator piping and it must be located at a point above the maximum expected level of the priming plus drain back/condensate water.

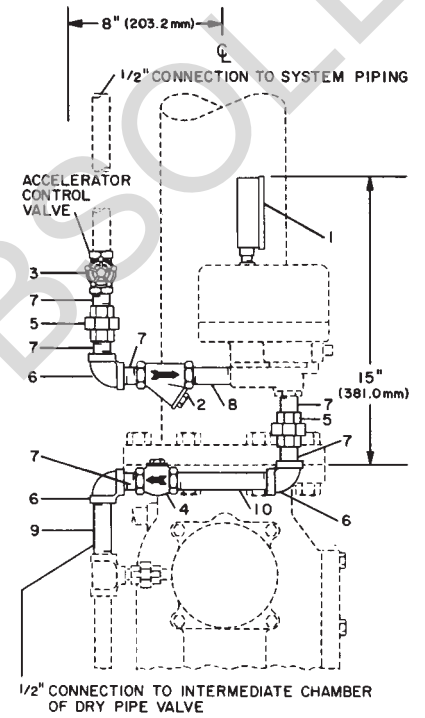
   If the connection is made to the riser, it must be located at least two feet above the level of the dry pipe valve priming water. Connections to a feed or cross main must be made either to the side or top of the main.

   NOTE

   Failure to follow the above instructions can result in flooding of the Accelerator and accidental tripping due to clogging of its Restriction.

5. An automatic drain, in accordance with the dry pipe valve manufacturer's specifications, must be located at the low point of the trim to the intermediate chamber.

6. The Accelerator/dry pipe valve combination must be installed in a heated enclosure which is maintained at a minimum temperature of 40°F/4°C. Heat tracing is not permitted.



1-250 lb. Air Pressure Gauge	6-1/2 90° Elbow (3 req'd)
2-1/2" Y-Strainer	7-1/2" x 1-1/2" Nipple (6 req'd)
3-1/2" Globe Valve	8-1/2" x 3" Nipple
4-1/2" Swing Type Check Valve	9-1/2" x 3-1/2" Nipple
5-1/2" Union (2 req'd)	10-1/2" x 5-1/2" Nipple

FIGURE E
INSTALLATION OF THE MODEL F311 ACCELERATOR

7. Close the Accelerator Control Valve.

   NOTE

   The Accelerator Control Valve must be closed during hydrostatic testing of the system, in order to prevent flooding of the Accelerator and clogging of the Restriction. Clogging of the Restriction can result in accidental tripping of the Accelerator.

   After the system has been hydrostatically tested and drained, the Accelerator connection to the system must be independently drained through the Strainer clean-out plug by first removing the Strainer clean-out plug and then opening the Accelerator Control Valve.

SETTING PROCEDURE

The Accelerator and dry pipe valve must be reset and restored to service as soon as possible after an operation. Follow the procedure indicated below.

a. Close the main control valve, the air supply control valve (to the system) and the Accelerator Control Valve.

b. Open the Inspector's Test Connection and then open the main drain valve as well as all auxiliary (low point) drains.

c. After the system has been drained, close the Inspector's Test Connection and all auxiliary drain valves. Leave the main drain valve open.

d. Set and prime the dry pipe valve in accordance with the instructions given in its Technical Data Sheet. Restore normal system air pressure. Leave the main control valve closed and the main drain valve open.

e. While holding the plunger of the Automatic Drain Valve depressed, open the Accelerator Control Valve one-quarter turn and allow the water in the Accelerator piping to blow out. After water spray stops discharging, close the Accelerator Control Valve and then release the plunger. *(This instruction does not apply when the F311 is being set for the first time, since the F311 is shipped in the set position. Proceed to Step 'f'.)*

f. Clean out the Strainer at the Accelerator Inlet.

   NOTE

   A clogged strainer can prevent the Accelerator from properly tripping the dry pipe valve.

g. Slowly remove the Vent Plug located in the front of the Accelerator Cover

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and bleed off any residual air pressure in the Differential Chamber.

h. Unscrew (counter-clockwise rotation) the knurled Reset Knob at the front of the Accelerator until it resists further turning. A click, which is the sound of the Lever snapping back into the Set Position, should be heard. Screw the Reset Knob back in until it is finger tight.

NOTE

Do not wrench on the Reset Knob. The Reset Knob will turn with finger torque only.

i. Replace the Vent Plug. *(Accelerators having a Serial Number less than 25000 utilized a Vent Plug with 1/4 inch pipe threads. In this case, a non-hardening type of pipe thread sealant must be applied sparingly to the male threads only.)*

j. Verify that the system air pressure has returned to normal.

k. Using a watch, note the time for the pressure in the Differential Chamber of the Accelerator to increase to 10 psi, after the Accelerator Control Valve is opened. The time should be within the range of values indicated below, for optimum performance of the Accelerator.

System Air Pressure, psi	Time to 10 psi, sec. Minimum	Maximum
20	24	160
25	18	116
30	15	92
40	10	60
50	8	48
60	6	36

NOTE

If the time to pressurize the Differential Chamber is not within the range of values given in the above table then the Accelerator Control Valve should be closed and the corrective procedure described in the Care and Maintenance Section followed.

l. When the air pressure in the Differential Chamber of the Accelerator is equal to that in the system, then the Accelerator is set and ready for service.

m. Close the Accelerator Control Valve and then slowly open the dry pipe valve's priming test valve, to bleed off any excess water above the priming level. Reclose the priming test valve, return system pressure to its normal value, and then re-open the Accelerator Control Valve.

n. Slowly open the main control valve. Close the main drain valve as soon as water discharges from the drain connection, and then completely open the main control valve. The fire protection system is now ready for service.

NOTE

After placing a fire protection system in service, notify the proper authorities and advise those responsible for monitoring proprietary and/or central station alarms.

CARE AND MAINTENANCE

The Model F311 Accelerator does not require any regularly scheduled maintenance. It is recommended, however, that its proper operation and condition be periodically verified in accordance with the following described Accelerator and System Inspection Procedures. Any impairment must be immediately corrected (refer to the Troubleshooting Section, as applicable).

It is recommended that automatic sprinkler systems be inspected by a qualified Inspection Service.

NOTES

If an Accelerator is to be temporarily taken out of service, then the proper authorities and all personnel who may be affected must be notified.

Before performing an alarm test, notify the proper authorities and all personnel who may be affected.

Before closing a fire protection system main control valve for inspection or maintenance work on the fire protection system which it controls, permission to shut down the affected fire protection system must be obtained from the proper authorities and all personnel who may be affected by this action must be notified.

Accelerator Inspection Procedure

It is recommended that the following Accelerator inspection procedure be performed at least annually, preferably in the fall or winter of the year. This procedure must also be used whenever flooding the system would expose the water to freezing conditions.

1. Verify that the Reset Knob is screwed in.

2. Close the system's main control valve and open the main drain valve to relieve the supply pressure to the dry pipe valve.

3. Verify that the Accelerator Control Valve is open.

4. Open the Inspector's Test Connection. Verify that the time to Accelerator trip is essentially the same as in previous tests. A momentary burst of air from the Automatic Drain Valve indicates that the Accelerator has tripped.

NOTE

As the system pressure is decreasing, check for any sign of water being discharged from the Accelerator Relief Port.

5. Depress the plunger of the Automatic Drain Valve. A steady stream of exhausting air indicates that the Accelerator has properly latched in the Tripped position.

6. Close the Accelerator Control Valve and the Inspector's Test Connection.

7. Clean out the Strainer at the Accelerator inlet.

NOTE

A clogged strainer can prevent the Accelerator from properly tripping the dry pipe valve.

8. Reset the Accelerator in accordance with Steps 'g' through 'n' of the Setting Procedure Section.

System Inspection Procedure

It is recommended that the following Accelerator and dry pipe valve inspection procedure be performed at least annually, preferably in the spring or summer of the year. This procedure can only be used whenever there is no danger that flooding the system will expose the water to freezing conditions.

1. Verify that the Reset Knob is screwed in.

2. Open the Inspector's Test Connection. Verify that tripping of the Accelerator operates the dry pipe valve and that water is delivered out of the Inspector's Test connection within the elapsed time required by the authority having jurisdiction. A maximum of 60 seconds (from the time that the Inspector's Test Connection is opened) is recommended.

NOTE

As the system pressure is decreasing, check for any sign of water being discharged from the Accelerator Relief Port.

3. Reset the Accelerator and dry pipe valve in accordance with the Setting Procedure Section.

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† DuPont Registered Trademark

TROUBLE SHOOTING

Refer to the following subsections, as applicable. If the designated instructions do no remedy the particular problem, refer to the Accelerator Disassembly and Reassembly subsection.

Water Discharge From Accelerator Relief Port

Follow the steps indicated below if water is discharged from the Accelerator Relief Port during a trip.

1. Verify that the connection from the Accelerator to the system piping is installed in accordance with Instruction No. 4 of the Installation Section. Correct if necessary.

2. Investigate for and correct any condition which could result in an excessive build-up of drain back and/or condensate water.

3. Review procedures which were used to set the Accelerator. Failure to perform Step 'e' of the Setting Procedure can permit a small amount of water to enter the Pilot Chamber of the Accelerator.

Slow Fill of Differential Chamber

Follow the steps indicated below, as required, if the time to fill the Differential Chamber is longer than the maximum value indicated in Step 'k' of the Setting Procedure Section.

1. Check to see that the Accelerator was reset per Step 'h' of the Setting Procedure Section.

2. Close the system's main control valve and open the main drain valve.

3. Check for any sign of external leakage past the Accelerator Pressure Gauge, Vent, and Restriction Access Plug connections.

4. Check for any sign of external leakage past the Reset Knob and Cover Gasket.

5. Close the Accelerator Control Valve.

6. Gently insert a 3/32 inch or smaller diameter probe up into the Relief Port. If the probe can be inserted more than 1/4 inch then the Lever has not reset and the Accelerator must be disassembled for internal inspection. See the instructions for Accelerator Disassembly and Reassembly.

7. Slowly remove the Accelerator Vent Plug to bleed off all pressure from the Differential Chamber and then slowly remove the Restriction Access Plug to bleed off all pressure from the Pilot Chamber.

8. Replace the Restriction and then the Restriction Access Plug. *(Accelerators having a Serial Number less than 25000 utilized a Restriction Access Plug with 1/2 inch pipe threads. In this case, a non-hardening type of pipe thread sealant must be applied sparingly to the male threads only.)*

9. Place the fire protection system back in service in accordance with Steps 'i' through 'n' of the Setting Procedure Section.

Unexplained Accelerator Trip

Follow the steps indicated below if there is an unexplained accidental trip of the Accelerator.

1. Verify that the connection from the Accelerator to the system piping is installed in accordance with Instruction No. 4 of the Installation Section. Correct if necessary.

2. Verify the time to fill the Differential Chamber as described in Step 'k' of the Setting Procedure Section. If the time to fill the Differential Chamber to 10 psi is longer than the maximum indicated value, then follow the instructions given under "Slow Fill of Differential Chamber".

3. If the time to fill the Differential Chamber is within the indicated range of values, then investigate for and correct any condition which could result in excessive leakage of system air pressure.

Fast Fill of Differential Chamber or Long Time to Accelerator Trip

Follow the steps indicated below, as required, if the time to fill the Differential Chamber is shorter than the minimum value indicated in Step 'k' of the Setting Procedure Section. This procedure should also be followed if the time to Accelerator trip (following opening of the Inspector's Test Connection) is significantly longer than expected.

1. Close the system's main control valve and open the main drain valve.

2. Close the Accelerator Control Valve.

3. Slowly remove the Accelerator Vent Plug to bleed off all pressure from the Differential Chamber and then slowly remove the Restriction Access Plug to bleed off all pressure from the Pilot Chamber.

4. Using a slotted screw driver, check the tightness of the Restriction.

5. Inspect the Restriction's O-ring seal. The O-ring must be replaced if there are any signs of nicks, cuts, or deterioration due to age. Replace the Restriction after cleaning and lubricating its O-ring with a non-petroleum based grease (such as Dow Corning FS3452). Replace the Vent Plug and Restriction Access Plug. *(Accelerators having a Serial Number less than 25000 utilized a Vent Plug and Restriction Access Plug with 1/4 and 1/2 inch pipe threads, respectively. In this case, a non-hardening type of pipe thread sealant must be applied sparingly to the male threads only.)*

6. If the Restriction and its O-ring were found to be in good condition, then it is likely that there is leakage past the Plunger. Remove the Cover from the Base. Check that the six screws securing the Upper Diaphragm Plate to the Cover are tight.

   Inspect the Upper Diaphragm for any sign of cracks, pin holes or deterioration due to age. Replace the Plunger if there is any possibility of leakage past the Upper Diaphragm.

7. Reassemble the Accelerator and place the fire protection system back in service in accordance with Steps 'j' through 'n' of the Setting Procedure Section.

Air Leakage Out Automatic Drain

If there is leakage of air out the Automatic Drain of the dry pipe valve after the Accelerator and dry pipe valve have been placed in service, then it will be necessary to first determine whether the leakage is past the Accelerator or the dry pipe valve.

Close the Accelerator Control Valve. Slowly remove the Accelerator Vent plug to bleed off all pressure from the Differential Chamber and then slowly remove the Restriction Access Plug to bleed off all pressure from the Pilot Chamber.

If leakage out the Automatic Drain persists then refer to the dry pipe valve Technical Data Sheet for maintenance instructions. If leakage out the Automatic Drain stops then the Accelerator will have to be taken out of service and the Accelerator Plug Seat removed for cleaning of the seat and the lower O-ring area on the Exhaust Valve.

Accelerator Disassembly and Reassembly (For Internal Inspection As Necessary)

NOTE

The following check-out procedure applies to Accelerators with Serial Numbers of 12,000 and greater. Refer to the 6-86 issue of TD109 for instructions

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applicable to Serial Numbers less than 12,000.

1. Close the system's main control valve and open the main drain valve.

2. Close the Accelerator Control Valve.

3. Slowly remove the Accelerator Vent Plug to bleed off all pressure from the Differential Chamber and then slowly remove the Restriction Access Plug to bleed off all pressure from the Pilot Chamber.

4. Break the union connections at the Inlet and Outlet of the Accelerator and remove it from the line. Plug the connection to the intermediate chamber of the dry pipe valve and place the fire protection system in service while the Accelerator is out for maintenance.

5. Remove the eight screws holding the Cover to the Base and remove the Cover.

6. Remove the six screws holding the Upper Diaphragm Plate to the Cover. Remove the Plunger and inspect the Upper Diaphragm to be sure that it is flexible and free from physical damage or deterioration due to age.

   Check the Jam Nut to assure that it is assembled tightly to its Screw. Remount the Plunger and Upper Diaphragm Plate taking care to cross-tighten the screws uniformly.

7. Replace the Restriction if it has been wetted. Clean and lubricate the Restriction O-ring seal with a non-petroleum based grease (such as Dow Corning FS3452).

   Replace the Vent Plug and Restriction access plug. *(Accelerators having a Serial Number less than 25000 utilized a Vent Plug and Restriction Access Plug with 1/4 and 1/2 inch pipe threads, respectively. In this case, a non-hardening type of pipe thread sealant must be applied sparingly to the male threads only.)*

8. Remove the Retaining Ring from the Upper Plug portion of the Exhaust Valve. Remove the four screws holding the Pivot Plate. Remove the sub-assembly of the Lever and Pivot Plate, the Exhaust Valve, the Anti-Flood Valve and the Relief Valve.

9. Inspect the Lower Diaphragm to be sure that it is flexible and free from physical damage or deterioration due to age.

10. Check to see that the Exhaust Valve components are securely assembled together. Only tighten by gripping the flats using an open end type wrench.

11. Inspect the O-ring on the Lower Plug. It must be replaced if there are any signs of nicks, cuts or deterioration.

12. Inspect the Relief and Anti-Flood Valves. If either is bent or nicked it must be replaced.

13. Remove and inspect the Latch. The catch leaf should extend out 5/16 to 3/8 inches in the free state.

14. Remove the Relief Valve Seat. Remove the O-ring and Seal Washer. Carefully clean the O-ring and Seal Washer seating surfaces in the valve seat and Accelerator Base. If the O-ring or Seal Washer is nicked, cut, or shows signs of deterioration, it must be replaced.

15. Replace the Seal Washer in the Accelerator Base. Apply a thin film of Dow Corning FS3452 Fluorosilicone Grease to the valve seat. Place the O-ring in its seat (the lubricant will hold it in place) and then thread the Relief Valve Seat into the Body with 10 to 15 ft.lbs. of torque.

16. Remove the Anti-Flood Seat Assembly w/Ball Float. Check for damaged parts and for freedom of Ball movement. If parts are damaged or inoperative, the Assembly must be replaced.

17. After checking the Anti-Flood Seat Assembly w/Ball Float, lubricate the O-ring with a thin film of Dow Corning FS3452 Fluorosilicone Grease, and thread the Assembly into the Body with 10 to 15 ft.lbs. of torque.

18. Remove the Reset Knob. Carefully clean the O-ring and its seating surface. If the O-ring is cut, nicked, or shows signs of deterioration it must be replaced. Lubricate the O-ring with a thin film of Dow Corning FS3452 Fluorosilicone Grease.

19. Reassemble the Accelerator in the following order.

    a. Thread the Reset Knob into the Base finger tight.

    b. Set the Anti-Flood Valve (with Retaining Ring in place) and the Compression Spring into their seat.

    c. Set the Exhaust Valve in place.

    d. Slide the Relief Valve into the slot at the end of the Lever and then remount the sub-assembly of the Lever and Pivot Plate in the Base, taking care to cross-tighten the screws uniformly.

    e. Replace the Retaining Ring on the Upper Plug.

    f. Push the Anti-Flood Valve end of the Lever down and release it twice to assure that there is no binding.

    g. Replace the Latch making sure that the notch in the bottom straddles the Reset Knob and that the tabs at the top are seated in the Base. Place the Lever in the Tripped (latched) position.

    h. Place the Cover upside down. Set the Cover Gasket in place and then push all eight screws through the Gasket to assist in assembling the Cover to the Base.

    i. Align the Cover with the Base and tighten all of the screws uniformly.

    j. Replace the Vent Plug and the Restriction Access Plug. *(Accelerators having a Serial Number less than 25000 utilized a Vent Plug and Restriction Access Plug with 1/4 and 1/2 inch pipe threads, respectively. In this case, a non-hardening type of pipe thread sealant must be applied sparingly to the male threads only.)*

    k. Reinstall the Accelerator and return the system to service in accordance with the Setting Procedure Section.

WARRANTY

Seller warrants for a period of one year from the date of shipment (warranty period) that the products furnished hereunder will be free from defects in material and workmanship.

For further details on Warranty, see Price List.

OBSOLETE

ORDERING PROCEDURE

Orders for F311 Accelerator, trim and replacement parts must include the description and Product Symbol Number (PSN). The Complete Model F311 Accelerator Package includes the Accelerator and Basic Trim.

Complete Package:
Specify: Complete Model F311 Accelerator Package, PSN 52-311-1-002.

Accelerator Only:
Specify: Model F311 Accelerator, PSN 52-311-1-001.

Basic Trim Only:
Specify: Basic Trim for Model F311 Accelerator, PSN 52-311-1-010.

Replacement Parts for Accelerator:
(Specify description) for use with Model F311 Accelerator, PSN (specify).

1-Base*	PSN 92-311-1-040
2-Cover	PSN 92-311-1-003
3-Upper Diaphragm Plate	PSN 92-311-1-009
4-Pivot Plate Assembly	PSN 92-311-1-011
5-Plunger	PSN 92-311-1-031
6-Exhaust Valve	PSN 92-311-1-008
6e-O-Ring	PSN 62-578-1-305
7-Rd.Hd. Mach. Screw 1/4" x 20UNC x 5/8"	PSN 62-672-1-150
8-Cover Gasket	PSN 92-311-1-007
9-Vent Plug**	PSN 18-313-3-005
10-O-Ring	PSN 62-578-1-304
11-Restriction	PSN 92-311-1-021
12-Restriction Access Plug**	PSN 18-313-3-007
13-Pan Hd. Mach. Screw No. 10-32UNF x 5/8"	PSN 62-679-1-108
14-Cotter Pin	PSN 62-380-9-070
15-Lever	PSN 92-311-1-012
16-Retaining Ring	PSN 62-937-1-006
17-Anti-Flood Valve*	PSN 92-311-1-035
18-Relief Valve	PSN 92-311-1-013
19-Spring*	PSN 92-311-1-033
20-Seat	PSN 92-311-1-016
21-O-Ring	PSN 62-561-1-006
22-Seal Washer	PSN 92-311-1-030
23-Latch	PSN 92-311-1-018
24-Fillister Hd. Mach. Screw 1/4" x 20UNC x 1-3/8"	PSN 62-678-1-165
25-Plug Seat	PSN 92-311-1-023
26-O-Ring	PSN 62-578-1-311
27-O-Ring	PSN 62-578-1-303
28-Reset Knob	PSN 92-311-1-017
29-Anti-Flood Seat Assembly w/Ball Float*	PSN 92-311-1-004
29f-O-Ring	PSN 62-578-1-314
-1.5 gms of FS3452	PSN 92-302-1-641
-2 oz. of FS3452	PSN 92-302-1-642

*Model F311 Accelerators manufactured prior to Serial Number 12000 utilized a Pilot Chamber Inlet Port configuration as shown in Figure F in lieu of the present Anti-Flood Seat Assembly w/Ball Float. To order the replacement parts shown in Figure F, specify: (specify description) for use with Model F311 Accelerator Serial Number (specify), PSN (specify). The Base for F311 Accelerators manufactured prior to Serial Number 12,000 are no longer available. The Base (PSN 92-311-1-040) is a direct replacement; however, the Anti-Flood Valve, Spring, and Seat Assembly w/Ball Float as listed in the present replacement parts list must also be ordered and installed.

**Model F311 Accelerators manufactured prior to Serial Number 25000 utilized a Vent Plug and Restriction Access Plug with 1/4 and 1/2 inch pipe threads, respectively.

Replacement Trim Parts:
Specify: (specify description), PSN (specify).

1/2" Swing Type Check Valve	PSN 46-049-1-004
1/2" Globe Valve	PSN 46-047-1-004
250 lb. Air Pressure Gauge	PSN 92-343-1-012
1/2" Strainer	PSN 52-353-1-005

PATENTS

The following patents are applicable to the Model F311 Accelerator:

COUNTRY	PATENT NO.
U.S.A.	4,570,719
United Kingdom	2,159,406

WEIGHTS

The nominal weight of the Complete F311 Accelerator Package is 14.7 pounds. The nominal weight of the Accelerator only is 6.8 pounds, and the nominal weight of the Basic Trim only is 8.1 pounds.

OBSOLETE

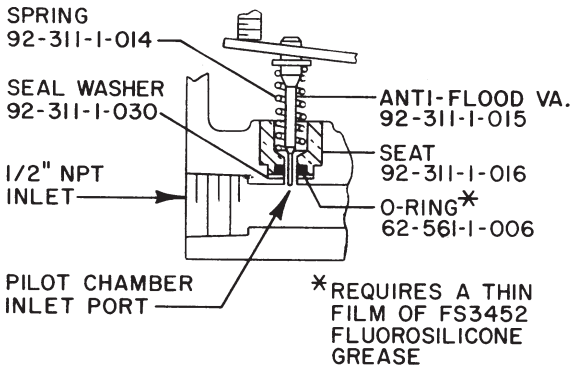


FIGURE F
PILOT CHAMBER INLET PORT
FOR F311 ACCELERATORS WITH SERIAL NUMBERS UP TO 12000